Time-Varying Covariates in Mortgage Survival

Part E(ii) — Andersen–Gill Cox Extension

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MTH 9877 — Baruch College, Spring 2026

May 13, 2026

Static Cox discards the most important prepayment signal.

Standard Cox (11 features)

One row per loan; all covariates fixed at origination. Macro features captured as vintage-era snapshots. Cannot track rate cycles, burnout, or equity build after origination.

Andersen–Gill (13 features)

One row per loan-month; $x_i(t)$ updated each period. Extends the same static base with the four time-varying features, tracking the full path of rate incentive and equity. 100K loans $\times$ ~47 months $=$ 4.68M intervals.

Time-varying features — updated every month

Feature	**What it captures**
`rate_incentive`	$r_{\text{orig}} - r_{\text{mkt}}(t)$: live option value to refinance; tracks rate cycles and burnout
`ELTV`	equity from amortisation and HPI gains; high ELTV suppresses refi
`unemployment`	macro labour cycle; proxies Fed easing in high-unemp periods
`hpi_yoy`	home-price appreciation unlocking cash-out refinancing

The Andersen–Gill model estimates $\boldsymbol{\beta}$ from monthly risk comparisons.

Hazard function (identical in form to standard Cox)

$$h_i(t) \;=\; h_0(t)\,\exp\!\big(\boldsymbol{\beta}^\top \mathrm{x}_i(t)\big)$$

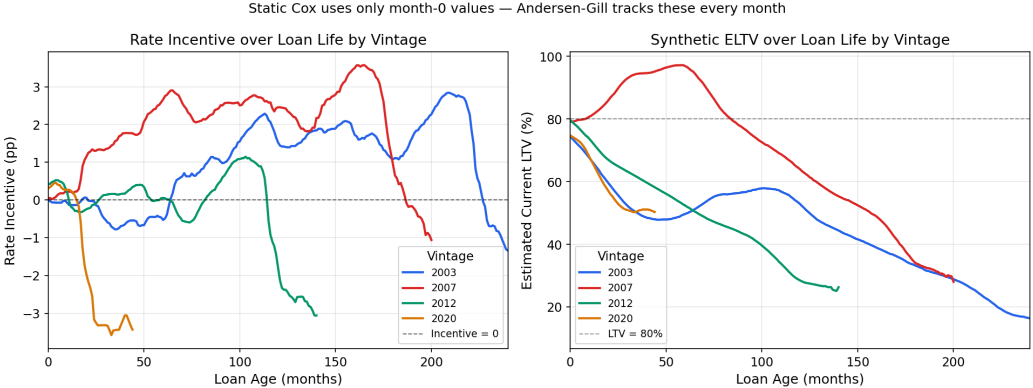
$\mathrm{x}_i(t)$ is the **current** covariate vector, updated each month. $\boldsymbol{\beta}$ is a single coefficient vector estimated across all loan-months.

Log partial likelihood (maximised to obtain $\hat{\boldsymbol{\beta}}$)

$$\ell(\boldsymbol{\beta}) \;=\; \sum_{i:\,\delta_i=1} \Bigg[\; \underbrace{\boldsymbol{\beta}^\top \mathrm{x}_i(t_i)}_{\substack{\text{log-score of}\\ \text{the prepaying loan}}} \;-\; \underbrace{\log \sum_{j\in\mathcal{R}(t_i)} \exp\!\big(\boldsymbol{\beta}^\top \mathrm{x}_j(t_i)\big)}_{\substack{\text{sum over all loan-months}\\ \text{active at } t_i}} \Bigg]$$

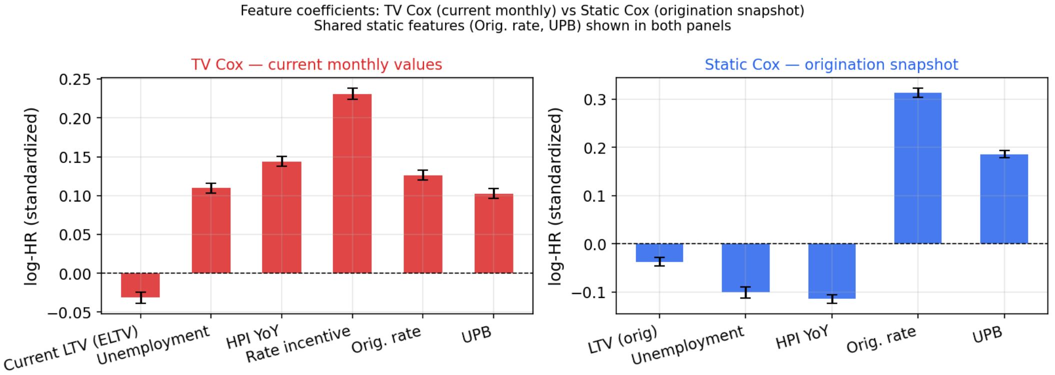
At each event: *given a prepayment at t_i, what is the probability it was loan i rather than any other active loan?* Maximising ℓ estimates $\hat{\boldsymbol{\beta}}$ without specifying $h_0(t)$.

Rate incentive and current LTV diverge sharply across vintages.



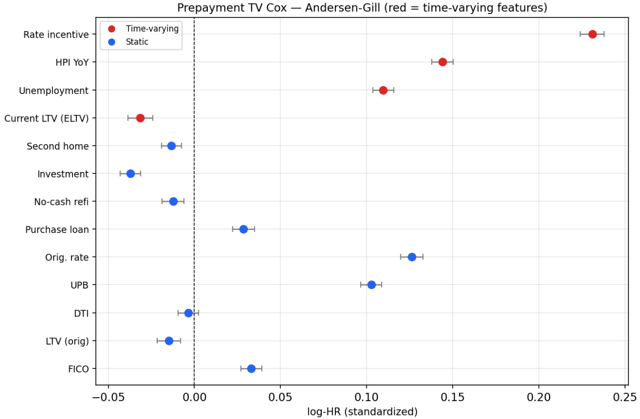
Static Cox uses only month-0 values — Andersen-Gill tracks these every month.

Live macro signals flip sign versus origination-era snapshots.



- `rate_incentive`: dominant positive predictor in TV Cox; near zero at origination, absent from Static Cox
- Macro signals (unemployment, HPI) flip sign — vintage-era snapshots capture a different regime than monthly values

TV Cox leads through $T = 60$; static Cox overtakes at $T = 84$–$T = 120$.



Landmark C-index

$$C(t^*) = P\big(\hat{h}_i(t^*) > \hat{h}_j(t^*) \mid T_i \le t^* < T_j\big)$$

Horizon	Static	**TV**	Δ
$T = 12$	0.609	**0.681**	$+0.072$
$T = 24$	0.627	**0.687**	$+0.060$
$T = 36$	0.630	**0.675**	$+0.045$
$T = 60$	0.625	**0.646**	$+0.021$
$T = 84$	**0.623**	0.605	-0.018
$T = 120$	**0.620**	0.583	-0.037

Key Takeaways

1. **Monthly-updated covariates add real information**
   Live features capture rate cycles and equity build invisible to the origination snapshot.
2. **TV Cox leads at short and medium horizons**
   $+0.072$ at $T = 12$, $+0.060$ at $T = 24$, $+0.045$ at $T = 36$, $+0.021$ at $T = 60$.
3. **TV Cox degrades beyond $T \approx 70$ months**
   Long-surviving loans are a selected group — their covariate state reflects why they haven't prepaid. A landmark super-model is required for long-horizon forecasting.
4. **Rate incentive is the key driver; inspect macro coefficients carefully**
   The live refi spread captures burnout and rate cycles. Unemployment flips sign due to multicollinearity with Fed easing.

Thank you

Questions?

Andersen & Gill (1982) *Ann. Stat.* · Cox (1972) *JRSS-B* · Sadhwani et al. (2021) *J. Fin. Econometrics*